MSG No. 177.149 $P622$ [Type I, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `622`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `622`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,3,5,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[2,4,6,7,8,9]`

Table 4: Wyckoff site: `2d`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[1,3,5,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[2,4,6,7,8,9]`

Table 5: Wyckoff site: `2e`, site symmetry: `6..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4,5,6]`
2	$[0, 0, -z]$	`[7,8,9,10,11,12]`

Table 6: Wyckoff site: `3f`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,11]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,5,9,10]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,6,8,12]`

Table 7: Wyckoff site: `3g`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,4,7,11]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,5,9,10]`

3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,8,12]`

Table 8: Wyckoff site: `4h`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,3,5]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[2,4,6]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[10,11,12]`

Table 9: Wyckoff site: `6i`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[2,5]`
3	$\left[0,\ \frac{1}{2},\ z\right]$	`[3,6]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,11]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[8,12]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[9,10]`

Table 10: Wyckoff site: `6j`, site symmetry: `.2.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7]`
2	$[x,\ x,\ 0]$	`[2,10]`
3	$[0,\ x,\ 0]$	`[3,8]`
4	$[-x,\ 0,\ 0]$	`[4,11]`
5	$[-x,\ -x,\ 0]$	`[5,9]`
6	$[0,\ -x,\ 0]$	`[6,12]`

Table 11: Wyckoff site: `6k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,7]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,10]`
3	$\left[0,\ x,\ \frac{1}{2}\right]$	`[3,8]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11]`
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[5,9]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,12]`

Table 12: Wyckoff site: 6l, site symmetry: ..2

No.	position	mapping
1	$[x, -x, 0]$	[1,12]
2	$[2x, x, 0]$	[2,7]
3	$[x, 2x, 0]$	[3,10]
4	$[-x, x, 0]$	[4,8]
5	$[-2x, -x, 0]$	[5,11]
6	$[-x, -2x, 0]$	[6,9]

Table 13: Wyckoff site: 6m, site symmetry: ..2

No.	position	mapping
1	$\left[x, -x, \frac{1}{2}\right]$	[1,12]
2	$\left[2x, x, \frac{1}{2}\right]$	[2,7]
3	$\left[x, 2x, \frac{1}{2}\right]$	[3,10]
4	$\left[-x, x, \frac{1}{2}\right]$	[4,8]
5	$\left[-2x, -x, \frac{1}{2}\right]$	[5,11]
6	$\left[-x, -2x, \frac{1}{2}\right]$	[6,9]

Table 14: Wyckoff site: 12n, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x-y, x, z]$	[2]
3	$[-y, x-y, z]$	[3]
4	$[-x, -y, z]$	[4]
5	$[-x+y, -x, z]$	[5]
6	$[y, -x+y, z]$	[6]
7	$[x-y, -y, -z]$	[7]
8	$[y, x, -z]$	[8]
9	$[-x, -x+y, -z]$	[9]
10	$[x, x-y, -z]$	[10]
11	$[-x+y, y, -z]$	[11]
12	$[-y, -x, -z]$	[12]